

EPT-TEST-2 (Full Length)

Question Paper Contain

50 Aptitude | 7 Algorithms | 7 C-Programming & DS

9 Maths | 8 CN | 8 DBMS | 6 OS | Digital 5

Total 100 Marks | 180 Minutes Duration

+1 Mark to each correct answer

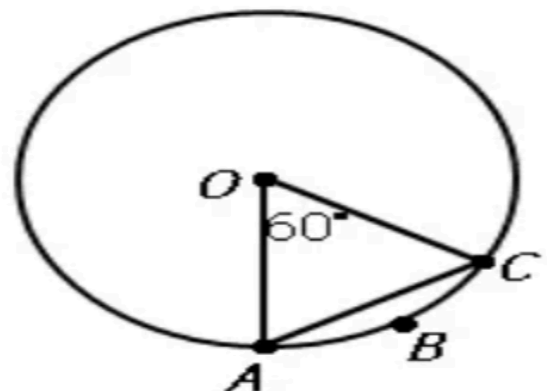
-0.25 Mark to each wrong answer

1.) How many unique ways can 4 identical green shirts and 2 identical red shirts be allocated among 6 children, ensuring each child gets a shirt?

- A. 15
- B. 40
- C. 216
- D. 720

2.) O serves as the midpoint of a circle having a radius of 6 units.

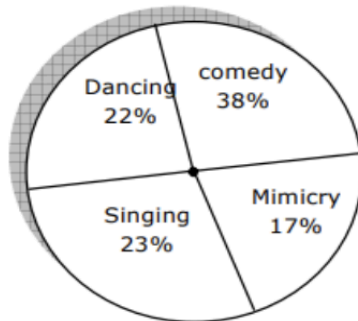
- A. The arc length of ABC surpasses 6 units.
- B. The arc length of ABC is below 6 units.
- C. The arc length of ABC equals 6 units.
- D. The correlation cannot be inferred from the provided data.



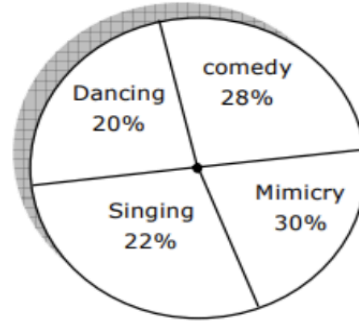
Q3.) 3. Percentage of employees engaged in various activities at Edu world:

What is the proportion of male participants to female participants in the dancing activity?

Total No. of Employees = 8000



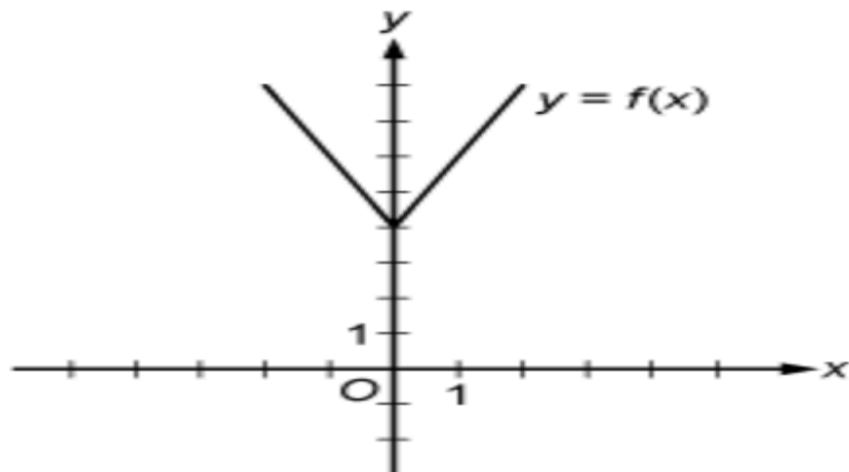
Total No. of Females = 3000



- A. 15:29
- B. 29:15
- C. 4:1
- D. 83:30

4.) The diagram below illustrates the plot of the function f defined as $f(x) = |2x| + 4$, valid for all values of x . (Note: $|2x|$ represents the absolute value of $2x$.) For which of the following functions g defined for all numbers x does the graph of g intersect the graph of f ?

- A. $g(x) = x - 2$
- B. $g(x) = x + 3$
- C. $g(x) = 2x + 3$
- D. $g(x) = 3x - 2$



5.) Below is a statement accompanied by two assumptions labeled I and II. An assumption refers to something taken for granted or presumed. Analyze the statements along with the given assumptions to determine which assumption(s) are implicitly suggested by the statement.

Statement: Despite a notable decrease in international crude oil prices, the Government has opted not to lower the prices of petroleum products.

Assumptions:

I. There might be a potential increase in crude oil prices on the global market soon.

II. The current price gap for petroleum products will enable the government to endure any potential price hikes in the future.

A If only Assumption I is implicit

B If only Assumption II is implicit

C If either Assumption I or II is implicit

D If neither Assumption I nor II is implicit

6.) Read the passage carefully and answer the question.

Nelson Mandela was designated as the national volunteer-in-chief for the Defense Campaign, with Maulvi Cachalia serving as his deputy. Cachalia's father had notably resisted alongside Gandhi in 1907. Mandela traveled across the Cape, Natal, and the Transvaal regions, visiting homes in townships to elucidate the plans, often engaging in discussions that lasted through the night. His mission was to instill confidence in people, encouraging them to confront oppression directly through non-violent means against the government. However,

typical challenges persisted due to racial discrimination in smaller towns, with limited facilities like hotels, taxis, and telephone lines available for Africans. This necessitated long walks to reach destinations and approaching potential hosts by knocking on doors. The reception varied, with some strangers showing enthusiasm while others were more cautious.

The primary aim of Mandela's discussions was to empower people

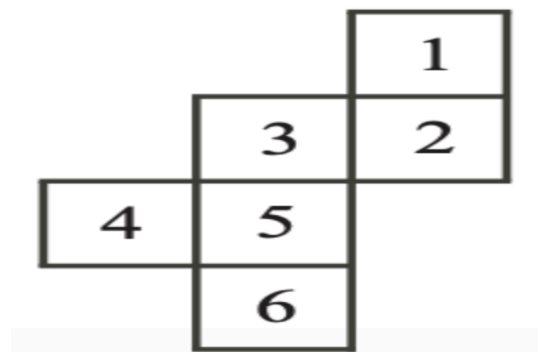
A. court arrests

B oppose oppression

C join him in his campaign tours

D get small town facilities

7.) The provided net is folded to create a cube. What is the result of multiplying the numbers on the four faces adjacent to the face labeled 1?



A. 120

B. 144

C. 180

D. 240

8.) A person moves 40 meters towards the south. Subsequently, turning right, he travels another 40 meters. He then takes a left and covers a distance of 20 meters. Finally, he turns left again and walks 40 meters. What is the distance of the man from his initial position now?

A. 60 meters

C. 100 meters

B. 40 meters

D. 80 meters

9.) Rafi mentioned to Mary, "I am thinking of watching a film this weekend." The statement in indirect speech would be:

Rafi told Mary that he _____ of watching a film that weekend.

- A. thought
- B. is thinking
- C. am thinking
- D. was thinking

10.) Permit: _____ : Enforce: Relax
(By word meaning)

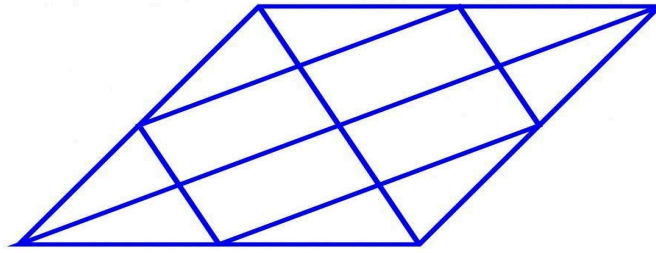
- A. Allow
- B. Forbid
- C. License
- D. Reinforce

11.) With a fair six-sided die labeled '1', '2', '3', '4', '5', and '6',
what is the likelihood of rolling a '1' on the initial roll and a '4' on the
subsequent roll?

- A. $1/36$
- B. $2/6$
- C. $5/36$
- D. $1/6$

12.) How many triangles are present in the given figure?

- A. 10
- B. 16
- C. 20
- D. 28



13.) A recent survey shows that 65% of tobacco users were advised to stop consuming tobacco. The survey also shows that 3 out of 10 tobacco users attempted to stop using tobacco.

Based only on the information in the above passage, which one of the following options can be logically inferred with *certainty*?

- A. A majority of tobacco users who were advised to stop consuming tobacco made an attempt to do so.
- B. A majority of tobacco users who were advised to stop consuming tobacco did not attempt to do so.
- C. Approximately 30% of tobacco users successfully stopped consuming tobacco.
- D. Approximately 65% of tobacco users successfully stopped consuming tobacco.

14.) The digit in the unit's place of the product $3^{999} \times 7^{999}$ is _____.

- A. 1
- B. 7
- C. 3
- D. 9

15.) "I cannot support this proposal. My _____ will not permit it."

- A. conscious
- B. consensus
- C. conscience
- D. consent

16.) Courts : _____ : : Parliament : Legislature (By word meaning)

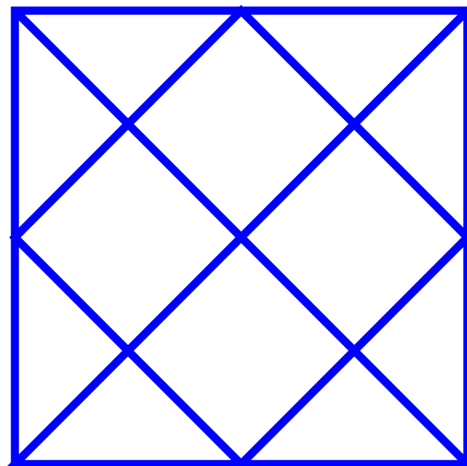
- A. Judiciary
- B. Executive
- C. Governmental
- D. Legal

17.) What is the smallest number with distinct digits whose digits add up to 36?

- A. 12355578
- B. 21345678
- C. 12345678
- D. 666666

18.) How many rectangles are present in the given figure?

- A. 8
- B. 9
- C. 10
- D. 12



19.) If $f(x)=x^2$ and $g(x)=\log_e(x)$, then $f(x)+g(x)$ will be

- A. Even function
- B. Odd function
- C. Both
- D. None of the above

20.) A can finish a task in 4 days. B requires twice the time taken by A. C needs twice the time taken by B, and D requires double the time taken by C to complete the same work. They form pairs, and one pair completes the work in two-thirds of the time taken by the other pair. Which pair comprises the first group?

- A. A, B
- B. A, C
- C. B, C
- D. A, D

21.) A college cricket team with 11 players consists of 4 batsman, 3 all-rounders, 3 bowlers and 1 wicket keeper. 4 players are selected randomly. Find the probability that the selection contains 2 batsmen, a bowler and an all-rounder.

- A. $54/330$
- B. $13/25$
- C. $12/55$
- D. $104/165$

22.) The expression $((x+y) - |x-y|)/2$ is equal to:

- A. The maximum of x and y
- B. The minimum of x and y
- C. 1
- D. None of the above

23.) Library: Book :: Book: _____

- A. Pages
- B. Copy
- C. Binding
- D. Cover

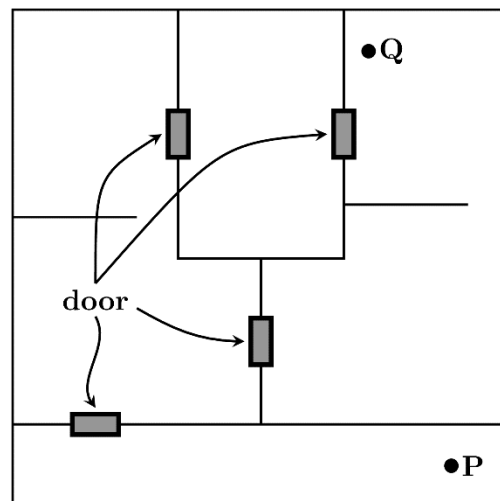
24.) Inhaling the smoke from a burning _____ could _____ you quickly.

- A. tire / tier
- B. tire / tyre
- C. tyre / tire
- D. tyre / tier

25.) A building has several rooms and doors as shown in the top view of the building given below. The doors are closed initially.

What is the minimum number of doors that need to be opened in order to go from the point P to the point Q?

- A. 4
- B. 3
- C. 2
- D. 1



26.) A game consists of spinning an arrow around a stationary disk as shown below. When the arrow comes to rest, there are eight equally likely outcomes. It could come to rest in any one of the sectors numbered 1,2,3,4,5,6,7 or 8 as shown.

Two such disks are used in a game where their arrows are independently spun.

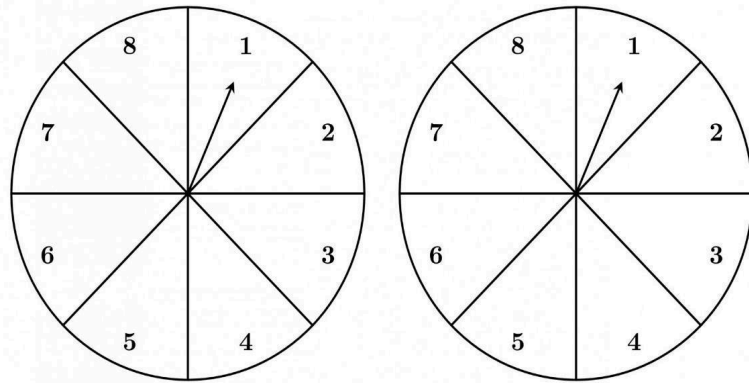
What is the probability that the sum of the numbers on the resulting sectors upon spinning the two disks is equal to 10 after the arrows come to rest?

A. $1/16$

B. $5/64$

C. $3/32$

D. $7/64$



27.) $x: y: z = 1/2:1/3:1/4$

What is the value of $(x+z+y)/y$?

A. 0.75

B. 1.25

C. 2.25

D. 3.25

28.) Consider the following equations of straight lines:

Line A: $2x-3y=5$

Line B: $3x+2y=8$

Line C: $4x-6y=5$

Line D: $6x-9y=6$

Which one among the following is the correct statement?

A. Line A is parallel to Line B and Line A is perpendicular to Line C

B. Line B is parallel to Line D and Line B is perpendicular to Line A

C. Line C is perpendicular to Line D and Line C is parallel to Line B

D. Line D is perpendicular to Line B and Line D is parallel to Line C

29.) Despite consistent underperformance, the chances of Hardik Pandya's inclusion in the MI lineup are _____

A. slim

B. bright

C. obvious

D. uncertain

30) Given the statements:

- P is the sister of Q.
- Q is the husband of R.
- R is the mother of S.
- T is the husband of P.

Based on the above information, T is _____ of S.

- A. the grandfather
- B. an uncle
- C. the father
- D. a brother

31.) Healthy eating is a critical component of healthy aging. When should one start eating healthy? It turns out that it is never too early. For example, babies who start eating healthy in the first year are more likely to have better overall health as they get older.

Which one of the following is the CORRECT logical inference based on the information in the above passage?

- A. Healthy eating is vital for those with optimal health conditions, but not for others.
- B. Beginning a healthy diet can be initiated at any age, with earlier being preferable.
- C. The connection between eating healthily and improved overall health is stronger during youth than later in life.
- D. Prioritizing healthy eating is more crucial for grown-ups than for children

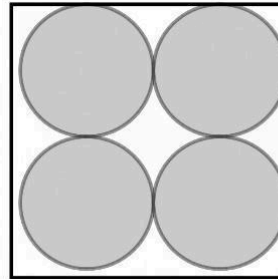
32.) For the past m days, the average daily production at a company was 100 units per day.

If today's production of 170 units changes the average to 110 units per day, what is the value of m ?

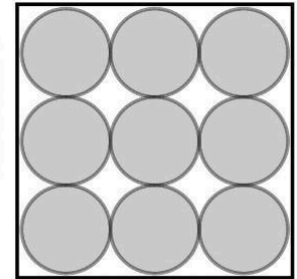
- A. 18
- B. 10
- C. 6
- D. 5

33.) In a square sheet of paper with 1 cm side length, case M has four shaded circles, and case N has nine shaded circles. What is the ratio of the unshaded areas in case M to that in case N?

- A. 2:3
- B. 1:1
- C. 3:2
- D. 2:1



Case M



Case N

34.) In a 12-hour clock operating accurately, how often do the hour, minute, and second hands align within a 12-hour period starting from 3 PM one day to 3 AM the following day?

- A. 11
- B. 12
- C. 144
- D. 2

35.) As you grow older, an injury to your _____ may take longer to _____.

- A. heel / heal
- B. heal / heel
- C. heal / heal
- D. heel / heal

36.) Three bells P, Q and R are rung periodically in a school. P is rung every 10 minutes; Q is rung every 20 minutes and R is rung every 30 minutes.

If all the three bells are rung at 12:00 PM, when will the three bells ring together again the next time?

- A. 1:00 PM
- B. 1:30 PM
- C. 2:00 PM
- D. 2:30 PM

37.) Below are two statements followed by four conclusions drawn from them: Statement 1: There exist bottles that are cups. Statement 2: Every cup is a knife.

Which of the following conclusions can be logically deduced from the above statements?

Conclusion I: Some bottles are knives. Conclusion II: Some knives are cups. Conclusion III: Every cup is a bottle. Conclusion IV: Every knife is a cup.?

- A. Only conclusion I and conclusion II are correct
- B. Only conclusion II and conclusion III are correct
- C. Only conclusion II and conclusion IV are correct
- D. Only conclusion III and conclusion IV are correct

38.) The price of an item is 10% cheaper in an online store S compared to the price at another online store M. Store S charges ₹ 150 for delivery. There are no delivery charges for orders from the store M. A person bought the items from the store S and saved ₹100.

What is the price of the item at the online store S (in ₹) if there are no other charges than what is described above?

- A. 2500
- B. 2250
- C. 1750
- D. 1500

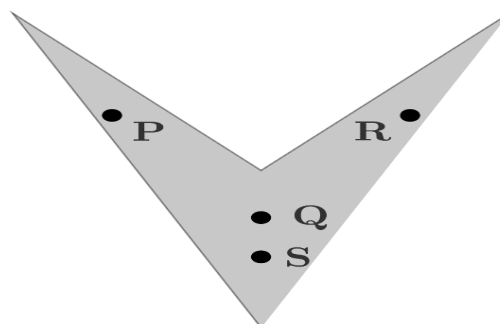
39.) Mosquitoes pose a threat to human health. Controlling mosquitoes using chemicals may have undesired consequences. In Florida, authorities have used genetically modified mosquitoes to control the overall mosquito population. It remains to be seen if this novel approach has unforeseen consequences.

Which one of the following is the correct logical inference based on the information in the above passage?

- A. Using chemicals to kill mosquitoes is better than using genetically modified mosquitoes because genetic engineering is dangerous
- B. Using genetically modified mosquitoes is better than using chemicals to kill mosquitoes because they do not have any side effects
- C. Both using genetically modified mosquitoes and chemicals have undesired consequences and can be dangerous
- D. Using chemicals to kill mosquitoes may have undesired consequences but it is not clear if using genetically modified mosquitoes has any negative consequence

40.) An art gallery engages a security guard to ensure that the items displayed are protected. The diagram below represents the plan of the gallery where the boundary walls are opaque. The location the security guard posted is identified such that all the inner space (shaded region in the plan) of the gallery is within the line of sight of the security guard.

If the security guard does not move around the posted location and has a 360° view, which of the following correctly represents the set of ALL possible locations

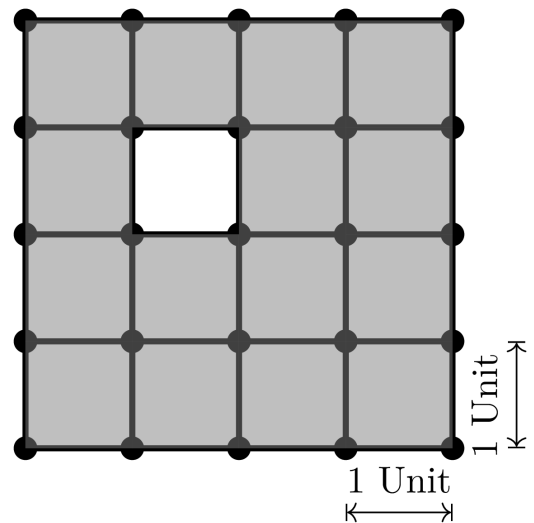


among the locations P, Q, R and S, where the security guard can be posted to watch over the entire inner space of the gallery.

- A. P and Q
- B. Q
- C. Q and S
- D. R and S

41.) The figure shows a grid formed by a collection of unit squares. The unshaded unit square in the grid represents a hole.

What is the maximum number of squares without a “hole in the interior” that can be formed within the 4×4 grid using the unit squares as building blocks?



- A. 15
- B. 20
- C. 21
- D. 26

42.) A is 3 times as efficient as B and A can finish a work in 40 days less than the time taken by B to finish the same work. In how many days can A and B together finish the work?

- A. 20
- B. 15
- C. 30
- D. 25

43.) A pond is inhabited by only four types of fishes, namely A, B, C and D. The number of Type A fishes is 25% more than that of type C; the number of type B fishes is 10% less than that of type A fishes, and the number of type D fishes constitutes 20% of the total fishes in the pond. If the number of type B fishes is 144, the total number of fishes in the pond is

- A. 500**
- B. 480**
- C. 540**
- D. 600**

44.) Fill in the gaps with the most appropriate word to complete the sentence.

Her complaints _____ with the complaints we have received from others.

- A agree**
- B similar**
- C identical**
- D tally**

45.) Choose the synonym for the word given below:

Vexation

- A Splendid**
- B Joyful**
- C Annoying**
- D Proud**

46.) $\log (x - 13) + 3 \log 2 = \log (3x + 1)$

A 20

B 21

C 22

D 24

47.) The following pie-chart represents the result of 600 successful students in various subjects of an examination.

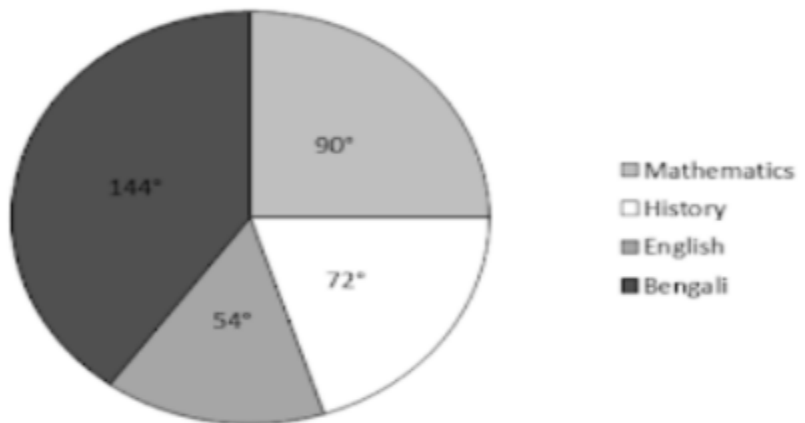
The percentage of students who passed in English is (English is 54 degree)

A 15%

B 20%

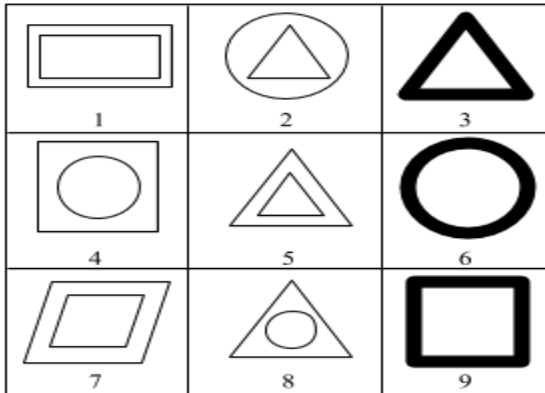
C 5%

D 12%



48.)

Group the given figures into three classes using each figure only once



- A. 1,5,7 ; 2,4,6 ; 3,9,8
- B. 1,4,7 ; 2,5,8 ; 3,6,9
- C. 1,5,7 ; 2,4,8 ; 3,6,9
- D. 1,7,9 ; 3,5,8 ; 2,4,6

49.) A fact that draws our attention is that, according to his position in life, an extravagant man is either admired or loathed. A successful business man does nothing to increase his popularity by being prudent with his money. A person who is wealthy is expected to lead a luxurious life and to be lavish with his hospitality. If he is not so, he is considered mean, and his reputation in business may even suffer in consequence. The paradox remains that he had not been careful with his money in the first place; he would never have achieved his present wealth.

The phrase 'lavish with his hospitality' in the third sentence of the passage above means

- A. Thoughtful in spending only on guests and strangers.
- B. Unconcerned in treating his friends and relatives.

C. Stinginess in dealing with his relatives.

D. Extravagance in entertaining guests.

50.) The difference between the sum of the first $2n$ natural numbers and the sum of the first n odd natural numbers is _____

A. $n^2 - n$

B. $n^2 + n$

C. $2n^2 - n$

D. $2n^2 + n$

51.) Consider the following matrices with these dimensions

A = 4×2

B = 2×2

C = 2×4

D = 4×3

The number of matrix scalar multiplication operations required for $((AB)(CD))$ _____.

A. 43

B. 64

C. 55

D. 56

52.)

```
#include <stdio.h>
```

```
int foo (int a)
```

```
{
```

```
    static int m=3;
```

```
    m = m+a;
```

```
    return m++;
```

```
}
```

```
int main()
```

```

{
  int i=0,j, a=10;
  for(j=1; a%j ==0; j++)
    i +=foo(a+j);
  printf("%d", i);
  return 0;
}

```

- A. 38
- B. 39
- C. 40
- D. 41

53.) Determine the highest possible number of edges in a planar bipartite graph with n vertices.

- A $3n + 2$
- B $4n + 6$
- C $2n - 4$
- D $3n - 6$

54.) If $f(n)=O(g(n))$, $g(n)=O(h(n))$ and $h(n)=\Omega(i(n))$, which of the following statements is TRUE?

- A $f(n)=O(i(n))$
- B $i(n)=O(f(n))$
- C $i(n)=O(h(n))$
- D $g(n)=O(i(n))$

55.) You are distributing IP addresses to devices within the 192.168.4.0 /26 network. Which IP addresses listed below can be allocated to devices in this subnet?

- A 192.168.4.0

- B 192.168.4.63
C 192.168.4.64
D 192.168.4.62

56.) Which statement correctly describes the characteristics of weak entities?

- A. Weak entities can be automatically removed when their associated strong entity is deleted.
B. Weak entities can be stored physically alongside their corresponding strong entities.
C. Both A and B are correct.
D. Neither A nor B is correct.

57.) Consider the following relation R with attribute A is primary key and B is the foreign key referring A with on delete cascade.

A	B
4	6
5	1
6	16
1	6
14	16
19	2
2	1
24	16
10	2
16	14

What is the number of tuples deleted additionally in order to preserve the referential integrity, when the tuple (1, 6) is deleted _____

- A. 1
B. 2
C. 3
D. 4

58.) The following is a dump of a UDP header in hexadecimal form:

06 32 00 0D 00 1D E2 17. What is the total length (in Bytes) of the UDP?

- A. 25
- B. 26
- C. 27
- D. 29

59.) An IP datagram carrying 10000 bytes of data with 20 bytes of the additional header must be sent over a link (i.e. network) that has an MTU of 4468 bytes. Assume the datagram has no Options. What will be the value of fragment size, MF for the second fragment?

- A. Fragment size = 4468, MF= 0
- B. Fragment size =5020, MF=1
- C.Fragment size= 5020, MF=0
- D.Fragment size = 4468, MF= 1

60.) In a client-server system using a MySQL database, the database server initially supports a maximum of X client requests. Later, this limit is increased by Y additional client requests. To accommodate the new limit using a counting semaphore S in the server design, what conditions should be satisfied?

- A. X
- B. Y

C. $X+Y$

D. $X-Y$

61.) What is a potential longest common subsequence for the given pair of strings: "ABABABA" and "BASSBAABA"?

A ABAABA

B BBABA

C BAABAA

D BAAAA

62.) Which statement from the following options is incorrect?

A. A port address directs a message to the appropriate application on a host.

B. The port number for POP3 is 100 and for DNS is 53.

C. A DNS client communicates with a DNS resolver to obtain the IP address of a domain.

D. IMAP offers greater functionality and features compared to POP3.

63.) A pouch holds 16 coins, out of which two are fake, having heads on both sides. The remaining coins are genuine. If a coin is chosen randomly from the pouch and flipped, what is the probability of obtaining a head?

A $9/16$

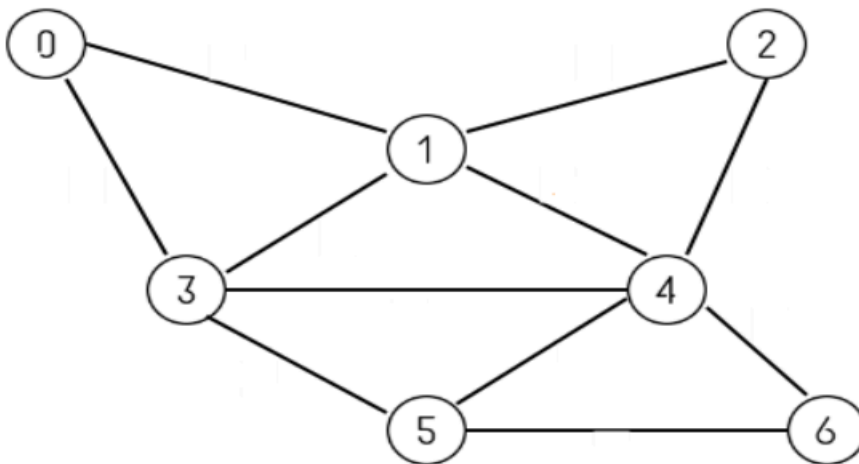
B $11/16$

C $5/9$

D None of the above

64.)

Consider the following weighted undirected graph



Which of the following is not a valid BFS Traversal

A 0, 1, 3, 2, 4, 5, 6

B 5, 3, 4, 6, 0, 1, 2

C 1, 0, 3, 5, 4, 2, 6

D 0, 3, 1, 5, 4, 2 ,6

65.) Two distinct numbers, x and y, are selected with replacement from the set {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10}. What is the probability that the absolute difference between x and y is greater than 5?

A 81/121

B 30/121

C 25/121

D 20/121

66.) The minimum number of literals for the complement of expression $(BC' + A'D)(AB' + CD')$ is_____

A. 0

B. 1

C. 2

D. 3

67.) #include <stdio.h>

int x = 5;

void m ();

void n ();

void main ()

{

int x=3;

m ();

printf ("%d", x);

}

void m ()

{

x = 8;

```

        n ();
    }
    void n ()
    {
        printf ("%d", x);
    }

```

A.38

B.83

C.20

D.40

68.) Consider the impact of slow start on a network with a round trip time of 10 msec. The initial congestion window size is 40 KB, and the sender's threshold is set to 20 KB. By the 9th transmission, a timeout occurs. After this timeout, how long does it take to fully transmit the first window of 16 KB? [Assume the window size at the beginning of the slow start phase is 2 KB.]

A 30 msec

B 40 msec

C 50 msec

D 60 msec

69.) The 2's complement representation of $(12121)_3$ is

A 01101000_2

B 10010111_2

C 01101011_2

D 01011011_2

70.) For each of the pair of Boolean formulae P and Q

I. $P = x \vee y, Q = x \wedge y$

II. $P = x \wedge (y \vee z), Q = (x \vee y) \wedge (x \vee z)$

III. $P = xy, Q = x\bar{y}$

Which of the following options is correct about the formula $P \oplus Q$?

- A.) I is Tautology
- B.) II is Tautology
- C.) III is Tautology
- D.) None of the above

71.) What is the block size of MD5?

- A 500 bits
- B 1012 bits
- C 1000 bits
- D 512 bits

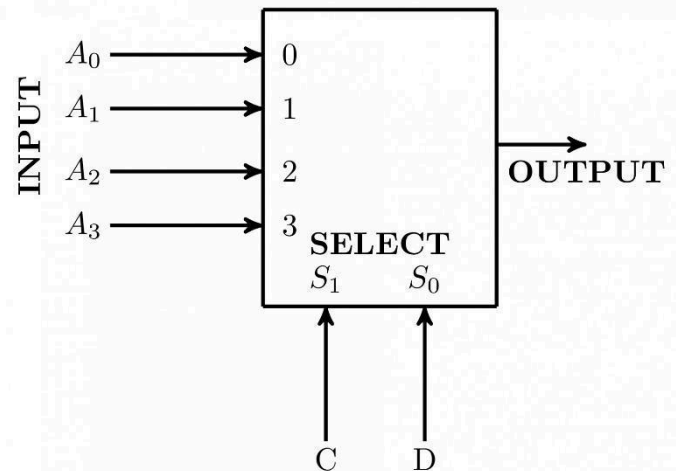
72.) Consider the following requests to read data from cylinders of a hard drive, where the current position of the head is at cylinder 73 and the head is moving towards cylinder 0. What is the order in which the cylinders in the queue would be serviced using the SCAN disk scheduling algorithm?

84,17,25,102,92,1,72,56,89,143,91,50,95,32

- A. 72,56,50,32,25,17,1,84,89,91,92,95,102,143
- B. 72,56,50,32,25,17,1,143,102,95,92,91,89,84
- C. 84,89,91,92,95,102,143,72,56,50,32,25,17,1
- D. 84,89,91,92,95,102,143,1,17,25,32,50,56,72

73.) Consider the 2-bit multiplexer (MUX) shown in the figure. For OUTPUT to be the XOR of C and D, the values for A₀, A₁, A₂ and A₃ are _____.

- A. A₀=0, A₁=0, A₂=1, A₃=1
- B. A₀=1, A₁=0, A₂=1, A₃=0
- C. A₀=0, A₁=1, A₂=1, A₃=0
- D. A₀=1, A₁=1, A₂=0, A₃=0



74.) The content of the registers are R1=25H, R2=30H and R3=40H. The following machine instructions are executed.

- PUSH {R1}
- PUSH {R2}
- PUSH {R3}
- POP {R1}
- POP {R2}
- POP {R3}

After execution, the content of registers R1,R2,R3R1,R2,R3 are

- A. R1=40H, R2=30H, R3=25H
- B. R1=25H, R2=30H, R3=40H
- C. R1=30H, R2=40H, R3=25H
- D. R1=40H, R2=25H, R3=30H

75.) Consider 4 characters a, b, c and d. Below are the corresponding codes to encode these characters.

$a=\langle 00 \rangle$, $c=\langle 01 \rangle$, $b=\langle 11 \rangle$, $d=\langle 101 \rangle$.

Which of the following option is true about given statements?

- S1: This is a prefix-free code.
- S2: This code could be generated by Huffman's algorithm.

- A. S1 is true but S2 is not true
B. S1 is not true but S2 is true
C. Both are true
D. Both are false

76.) Consider the following two systems of linear equations :

S1:

$$x_1 + x_2 = 10$$

$$-x_1 + x_2 = 0$$

S2:

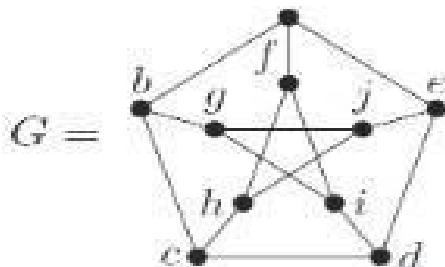
$$x_1 - 2x_2 = -3$$

$$2x_1 - 4x_2 = 8$$

Which of the above systems has infinite solutions?

- A. Only S1
B. Only S2
C. Both
D. None

77



The chromatic number for the above graph G is

- A. 2
- B. 3
- C. 4
- D. 5

78.) $\int \frac{x+1}{\sqrt{x^2+2x+3}} =$

- A $\sqrt{x^2 + 2x + 3} + c$
- B $2\sqrt{x^2 + 2x + 3} + c$
- C $-\sqrt{x^2 + 2x + 3} + c$
- D None of the above

79.) In the quicksort of the following numbers, if the pivot is chosen as the first element, what will be the orders of the numbers after the use of partition function? assume we are sorting increasing order

11,15,9,13,17,7,5,12,6,18

- A 6 7 9 5 11 15 13 12 17 18
- B 6 9 7 5 11 15 13 17 12 18
- C.6 9 7 5 11 15 13 12 17 18

D. 6 5 8 7 11 15 13 12 17 18

80.) For a given relation having 7 attributes $A_1, A_2, A_3, A_4, A_5, A_6$. and A_7 , the candidate keys are $\{A_1, A_2, A_6\}$. The number of super keys possible are

- A. 112**
- B. 100**
- C. 64**
- D. 80**

81.) Consider the statements below about scheduling algorithms in operating systems.

- I. If all jobs have identical run lengths, a RR (Round Robin) scheduler (with a time-slice much shorter than the jobs' run lengths) provides better average turnaround time than FIFO.**
- II. We can practically implement a scheduling algorithm that achieves the optimal average waiting time.**
- III. Round-robin scheduling ensures no starvation.**

Which of the following options is correct?

- A. I and II are TRUE but III is False**
- B. I is TRUE but II and III are False**
- C. I and II are False but III is True**
- D. All statements are true**

82.) Consider a computer system equipped with a hard disk for data storage. The responsibility of overseeing the hard disk falls under the jurisdiction of the operating system. It's crucial to monitor the utilization of the hard disk, determining which sectors are available and which are already in use. To manage this, a specific disk component employs a binary string to indicate the status of its sectors: 0 signifies an available sector, while 1 indicates an occupied sector. A 32-bit portion of this binary string has a hexadecimal representation of F1E0FA36. The approximate percentage of unoccupied sectors for this segment of the disk is _____

- A. 40.50
- B. 43.75
- C. 45.25
- D. 48.50

83.) What will be printed by following the C code?

```
int a[7] = {0, 1, 2, 3, 4, 5, 6};  
int *p = &a[3];  
p += 2;  
*p += 2;  
printf ("%d", *p++);
```

- A. 6
- B. 7
- C. 8
- D. 9

84.) During normal IP packet forwarding by a router, which of the following packet fields are updated?

- 1. Destination address**
- 2. Port address**
- 3. Checksum**
- 4. TTL**

A All the mentioned fields

B Only 2, 3, 4

C Only 1, 2

D Only 3, 4

85.) What will be the output of the following c code?

```
#include <stdio.h>
int main(void) {
    int i=1;
    if(i==1) {
        i=((7, (i=4)),i=2);
        printf ("%d", i);
    }
    else
        printf("Hello");
    return 0;
}
```

A. 4

B. 2

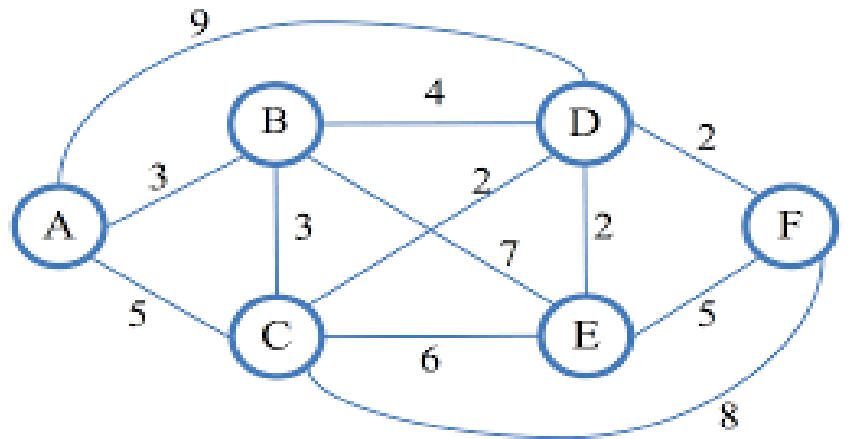
- C. 7
- D. Hello

86.) The probability that an event A happens in one trial of an experiment is 0.4. Three independent trials of the experiment are performed. The probability that the event A happens at least once

- A.0.936
- B.0.784
- C 0.904
- D None of the above

87.) The cost of the MST of the below graph is given by

- A. 10
- B. 11
- C. 12
- D. 13



88.) In a circular doubly-linked list where only a head pointer is maintained, pointing to the initial node, which of the subsequent operations can be executed in constant time ($O(1)$)? [Note: We possess solely the head pointer and a pointer to the node being inserted.]

- i) Addition at the beginning of the linked list
- ii) Addition at the tail of the linked list
- iii) Removal of the initial node of the linked list
- iv) Removal of the terminal node of the linked list

A Only I, II

B Only II and III

C Only I, II, III

D All four statements

89.) Simplified Boolean expression for $A'BC + AB'C' + A'B'C' + AB'C + ABC$

A. AB

B. $B'C$

C. $AB + (A' + AB')C$

D. $AB' + BC + B'C'$

90.) Given that $(292)_{10} = (1204)_x$ in some number system x . The base x of that number system is

A. 2

B. 8

C. 10

D. None of the above

91.) Consider the following relation $R(A, B, C, D, E)$ and functional dependencies F that hold over this relation.

$ABD \rightarrow C,$

$BC \rightarrow D,$

$CD \rightarrow E$

In which normal form is relation R ?

A. 1NF

B. 2NF

C. 3NF

D. BCNF

92.) Given the following relation instances

X	Y	Z
1	4	2
1	5	3
1	4	3
1	5	2
3	2	1

Which of the following conditions is/are TRUE?

- A. $XY \rightarrow Z$
- B. $YZ \rightarrow X$
- C. $X \rightarrow Y$
- D. $XZ \rightarrow Y$

93.)

1. Unique not null is equivalent to primary key.
2. Relational Algebra and SQL have the same expressive power.

Which of the above statements are False?

- A. 1
- B. 2
- C. Both
- D. None

94.) For the following relational algebra expressions , the relation schemas are R(a, b) and S(b, c)

Q1: $R \bowtie S$

Q2: $\sigma_{(R.b = S.b)}(R \times S)$

Which of the following is true about the given queries Q1 and Q2?

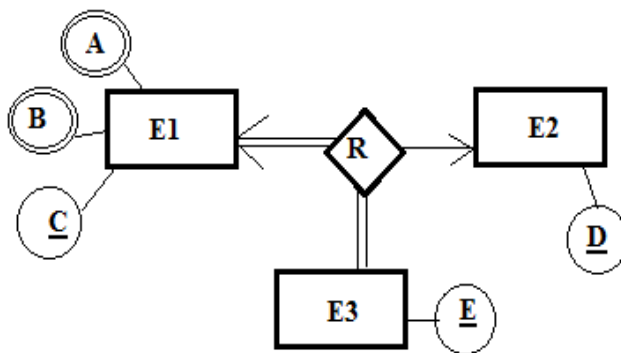
A Q1 and Q2 produce the same answer.

B The answer to Q1 is always contained in the answer to Q2.

C The answer to Q2 is always contained in the answer to Q1.

D Q1 and Q2 produce different answers.

95 The minimum number of tables required for the above ER diagram which satisfy 1NF are _____



A. 2

B. 3

C. 4

D. 5

96.)

$$A = \begin{bmatrix} 2000 & 2002 & 2004 \\ 2005 & 2007 & 2009 \\ 2008 & 2010 & 2012 \end{bmatrix}$$

- A. 0
- B. 1
- C. 2
- D. 3

97.)

A system has 4 processes and 5 allocatable resource. The current allocation and maximum needs are as follows:

	Allocated	Maximum	Available
Process A	1 0 2 1 1	1 1 2 1 2	0 0 x 1 1
Process B	2 0 1 1 0	2 2 2 1 0	
Process C	1 1 0 1 0	2 1 3 1 0	
Process D	1 1 1 1 0	1 1 2 2 1	

The smallest value of x for which this is a safe state is ____.

- A. 0
- B. 1
- C. 2
- D. 3

98.) Consider a system with a single processor unit. In this system suppose a new process arrives at an average of 10 processes per

minute and each such process requires an average of 5 seconds of service time. Which of the following is the fraction of time (in %) CPU is idle

- A 66.67 %
- B 33.33 %
- C 20%
- D 16.67 %

99.) There is one stack s and one queue q and then following operations are performed upon these two.

Push A, enqueue B, push C, pop, pop, enqueue D, push E, dequeue, enqueue F

Then choose the correct option for the total number of elements, which are left in the queue and stack in the end.

- A 6
- B 5
- C 4
- D 3

100.) In an IPV4 packet, the value of MF = 1 and the MTU of the network is 500 Bytes. The header size is 20 Bytes. The packet size is big enough so we need to divide it into multiple fragments. If the fragment offset of some packet is 180 then what will be the fragment number?

- A 1
- B 2
- C 3
- D 4

1.) SOLUTION A

Ans is permutation of GGGGRR which is $6! / 4!2! = 15$

2.) SOLUTION A.

Given that the radius measures 6 units, it follows that both OA and OC are also 6 units in length. Consequently, angles OAC and OCA are equal. This implies that all angles within triangle OAC measure 60 degrees.

The length of segment AC is determined to be 6 units.

It's essential to note that the shortest distance between two points is a straight line connecting them. As a result, the arc length of AC is greater than the length of the straight-line segment AC

3.) SOLUTION B

The number of employees who took part in dancing is calculated as 22% of 8,000, which equals 1,760 individuals.

Out of the total participants, the count of female employees participating in dancing stands at 20% of 3,000, amounting to 600 women.

To determine the number of male participants, we subtract the count of female participants from the total participants, resulting in 1,160 male employees.

Thus, the ratio of male to female participants can be expressed as 1,160 to 600, which simplifies to 29 to 15.

4.) SOLUTION D

Take Option D :

Put $g(x)=3x-2=|2x|+4$

If $x<0$ then $3x-2=-2x+4$

So, $5x=6$, So, $x=6/5$ which is contradicts that $x<0$

So, if $x\geq 0$ then $3x-2=2x+4$

$x=6$, which is correct.

So, $g(x)=3x-2$ intersects with $f(x)$ at point $x=6$.

5.) Solution: D

The government's decision-making might not always consider the possibility of a price increase. In reality, our oil companies are experiencing losses even with the decrease in global prices. Therefore, both assumptions are not valid

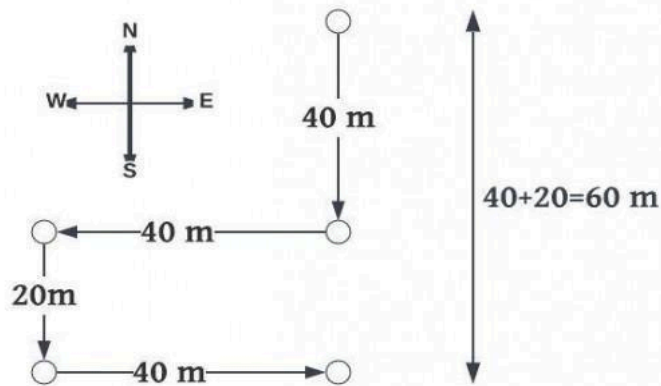
6.) Solution B

Mandela delivered speeches to motivate and gain the trust of the people, aiming to combat oppression. This information is mentioned in the fifth line of the text.

7.) SOLUTION B

Answer B is 144. When the provided net is folded, face number 5 will align opposite to face number 1. Consequently, the other four faces connect to face number 1 along an edge. Thus, the multiplication of these numbers results in $2 \times 3 \times 4 \times 6$, which equals 144.

8.) SOLUTION A



9.) SOLUTION D

In indirect narration, pronouns and tenses are modified. "I" transforms into "he", and "am thinking" adjusts to "was thinking", moving the tense one step backward to the past. The temporal phrase "this weekend" is altered to "that weekend" to maintain consistency in the shift from direct to indirect speech.

Rafi told Mary that he was thinking of watching a film that weekend

10.) SOLUTION B

The analogy demonstrates opposing relationships. "Permit" contrasts with "Forbid", similar to how "Enforce" opposes "Relax".

"Permit" signifies granting permission or allowing.

"Forbid" indicates prohibiting or disallowing.

"Enforce" refers to ensuring adherence to a regulation or law.

"Relax" denotes being in a state of ease. Correct Answer: B

11.) Solution A

We possess a fair six-sided die. Upon rolling:

- **There's an equal chance of 1 in 6 (or a $1/6$ probability) to obtain a '1' on the first attempt.**
- **There's an equivalent chance of 1 in 6 (or a $1/6$ probability) to secure a '4' on the second attempt.**

Given that these rolls are independent, with one roll not influencing the outcome of the other, we can calculate the combined probability by multiplying the individual probabilities:

$$(1/6) * (1/6) = 1/36$$

Hence, the probability of obtaining a '1' on the first roll followed by a '4' on the second roll is $1/36$.

12.) SOLUTION

Ans c 20.

13.) Solution B

The passage provides two pieces of information:

- **65% of tobacco users were advised to stop consuming tobacco.**
- **3 out of 10 tobacco users attempted to stop using tobacco.**

Now, let's consider these percentages in the context of the question:

If 65% of tobacco users were advised to stop, it means that 35% were not advised to stop because the total percentage adds up to 100%.

If 3 out of 10 tobacco users attempted to stop, it means 7 out of 10 (or 70%) did not attempt to stop.

Given this information, we can make the following logical inference:

A majority of tobacco users who were advised to stop consuming tobacco did not attempt to do so.

This is because 70% (the percentage who did not attempt to stop) is greater than 50% which means that the majority did not make an attempt to stop.

14.) SOLUTION A

The pattern of the unit's digits of powers of 3:

- 1. $3^1=3$**
- 2. $3^2=9$**
- 3. $3^3=7$**
- 4. $3^4=1$**

The pattern repeats every 4 powers of 3.

The pattern of the unit's digits of powers of 7:

- 1. $7^1=7$**
- 2. $7^2=9$**
- 3. $7^3=3$**
- 4. $7^4=1$**

The pattern repeats every 4 powers of 7.

Now, Unit digit of $3^{999} \times 7^{999} = (3^{4(249)} \times 3^3) \times 7^{4(249)} \times 7^3 = 1 \times 7 \times 3 = 21$

So, the unit's digit of $3^{999} \times 7^{1000}$ is 7.

15.) SOLUTION C

In the context provided, the most suitable word for the blank is "conscience." Let's examine the definitions of the other choices:

Conscious: This term signifies being aware or having knowledge of something. However, in the given sentence, it doesn't convey the concept of moral judgment or an internal guide, which is what the sentence is implying.

Consensus: This word refers to general agreement or harmony among a group. While it relates to decision-making, it doesn't align with the phrase "will not permit it," which indicates an individual's personal conviction rather than a collective agreement.

Conscience: This term denotes an individual's inner sense of right and wrong that guides their actions and choices. It serves as an individual's moral compass and is the most fitting word for the sentence. It suggests that the speaker's personal moral beliefs or values prevent them from endorsing the proposal.

Consent: This word means giving approval or permission for something. While it indicates permission, it doesn't encompass the idea of personal moral judgment that the sentence is trying to convey. Therefore, the correct word choice for the sentence is "conscience." The revised statement would be: "I cannot support this proposal. My conscience will not permit

16.) SOLUTION A

The analogy can be interpreted in the following manner: Courts relate to Judiciary in the same way as Parliament relates to Legislature.

In this analogy, the connection is rooted in the meanings of the words and the respective roles and responsibilities they represent. The Judiciary is the governmental branch responsible for interpreting and applying laws, ensuring justice and settling legal conflicts. Likewise, the Legislature is the governmental branch tasked with formulating and enacting laws, representing the populace's interests.

Hence, the term “Judiciary” is the correct selection to fill in the analogy.

17.) SOLUTION C

To find the smallest number with distinct digits whose digits add up to 36, we need to consider the given options and determine which one satisfies the conditions.

Let's analyze the options provided:

- A. 12355578: The sum of the digits in this number is $1+2+3+5+5+5+7+8=36$. However, this number contains repeating digits (three 5s), which contradicts the requirement of distinct digits.
- B. 21345678: The sum of the digits in this number is $2+1+3+4+5+6+7+8=36$. Additionally, all the digits in this number are distinct, which satisfies the condition of distinct digits. However, we need to find the smallest number that meets the criteria.

- C. 12345678: The sum of the digits in this number is $1+2+3+4+5+6+7+8=36$. Moreover, all the digits in this number are distinct. Importantly, this is the smallest arrangement of distinct digits that adds up to 36. Each digit from 1 to 9 is used exactly once, and there is no smaller number with distinct digits that satisfies the condition.
- D. 666666: The sum of the digits in this number is $6+6+6+6+6+6=36$. However, this number contains repeated digits (six 6s), which does not satisfy the requirement of distinct digits.

Based on the analysis, the correct answer is 12345678. It is the smallest number with distinct digits whose digits add up to 36.

Short Method: Options A and D are not valid choices as they both have repeated digits. On the other hand, when comparing options B and C, it becomes evident that 12345678 is indeed the smallest number with distinct digits whose digits add up to 36.

18.) SOLUTION C

Ans c 10

19.) SOLUTION D

since $f(x) = x^2$ is an even function and $g(x)=\log_e(x)$ is undefined

hence $f(x) + g(x)$ would also be undefined, so the correct option would be D.

An even function adheres to the principle: $f(-x) = f(x)$ for every x within the function's domain.

Conversely, an odd function follows the guideline: $f(-x) = -f(x)$ for every x within the function's domain.

20.) SOLUTION D

A can accomplish $\frac{1}{4}$ of the task in a day.

B can complete $\frac{1}{8}$ of the task in a day.

C can finish $\frac{1}{16}$ of the task in a day.

D can achieve $\frac{1}{32}$ of the task in a day.

The combined daily work of B and C is $\frac{1}{8} + \frac{1}{16} = \frac{3}{16}$.

The combined daily work of A and D is $\frac{1}{4} + \frac{1}{32} = \frac{9}{32}$.

Thus, A and D can complete the task in $\frac{32}{9}$ days, which is equivalent to $\frac{2}{3}$ of the time taken by B and D, who complete it in $\frac{16}{3}$ days.

Therefore, the correct pair is A and D. Option D.

21.) SOLUTION A

$$(4C2 * 3C1 * 3C1) / 11C4$$

$$= 54/330$$

22.) Solution B

When $x > y$, $|x - y| = x - y$, if we substitute in expression we get y .

When $x < y$, $|x - y| = -(x - y)$, if we substitute in expression we get x .

Therefore in both the case we get minimum of (x, y) .

ANS: B

23.) SOLUTION A

Answer Option (A) is correct.

A library consists of books, in the same way, books have pages.

24.) SOLUTION C

Tyre: A circular covering for a wheel.

Tier: Series or levels of something organized in increasing order.

Tire: To become fatigued or drained of energy from physical exertion..

Inhaling the smoke from a burning tyre could tire you quickly.

25.) SOLUTION:C

26.) SOLUTION:D

When we select only one number from any of the two disks, then the probability $=1/8$

When we select one number from the first disk and the one number from the second disk simultaneously, then the probability $=1/8 \times 1/8 = 1/64$

The numbers on the resulting sectors upon spinning the two disks is equal to 8 after the arrows come to

rest $= (2,8), (3,7), (4,6), (5,5), (6,4), (7,3), (8,2)$ _٧ Number of favorable outcomes = 7

$\therefore$ The required probability $= 7/64$

27.) SOLUTION:D

Given that $=1/2:1/3:1/4$

$$\Rightarrow x:y:z = (1/2 \times 12):(1/3 \times 12):(1/4 \times 12) [\because \text{LCM}(2,3,4) = 12]$$

$$\Rightarrow x:y:z = 6:4:3 = k(\text{let})$$

$$\Rightarrow x=6k, y=4k, z=3k$$

$$\therefore \text{The value of } (x+z-y)/y = (6k+3k+4k)/4k = 13k/4k = 3.25.$$

28.) Solution D

Given the system of equations:

1. $2x-3y=5$

2. $3x+2y=8$

3. $4x-6y=5$

4. $6x-9y=6$

From equation (1): $2x-3y=5 \Rightarrow -3y=-2x+5 \Rightarrow y=2/3x-5/3$ Slope of the line, $m_1=2/3$

From equation (2): $3x+2y=8 \Rightarrow 2y=-3x+8 \Rightarrow y=-3/2x+4$ Slope of the line, $m_2=-3/2$

From equation (3): $4x-6y=5 \Rightarrow -6y=-4x+5 \Rightarrow y=2/3x-5/6$ Slope of the line, $m_3=2/3$

From equation (4): $6x-9y=6 \Rightarrow -9y=-6x+6 \Rightarrow y=2/3x-2/3$ Slope of the line, $m_4=2/3$

We find that $m_1=m_3=m_4=2/3$. This implies that lines *A*, *C*, and *D* are parallel to each other.

Further, $m_1 \times m_2 = 2/3 \times (-3/2) = -1$

$m_3 \times m_2 = 2/3 \times (-3/2) = -1$

$m_4 \times m_2 = 2/3 \times (-3/2) = -1$

This shows that lines *A*, *C*, and *D* are perpendicular to *B*

29.) SOLUTION C

First and Second sentences are contradictory as *Despite* is being used in the first statement so the chance of selection must be very high. We can eliminate option A and D. Now we can not use C with the word chances so B is the correct option.

30.) SOLUTION B

T is an uncle of S.

31.) SOLUTION B

The passage underscores the idea that healthy eating knows no age boundaries, advocating that it's never too early to begin. It highlights the advantageous impact of early healthy eating habits on long-term health, even citing infants who start eating healthily in their first year as likely to experience improved overall health as they grow older.

Hence, the logical inference derived from this is that healthy eating can commence at any stage of life, with an emphasis on the earlier adoption of such habits for optimal health as individuals age.

32.) SOLUTION C

Given that, for the past m days, the average daily production at a company was 100 units per day.

- 1day $\rightarrow$ 100units
- m days $\rightarrow$ 100 units

If today's production of 180 units changes the average to 110 units per day.

$$\text{Now, } 100m+170 = 110(m+1)$$

$$\Rightarrow 100m+170=110m+110$$

$$\Rightarrow 10m=60$$

$$\Rightarrow m=6$$

33.) SOLUTION B

Let's denote the radius of the larger circle as R cm and the radius of the smaller circle as r cm.

From Case M and Case N, we derive:

- $R+R+R+R=1$ which gives $4R=1$ and thus $R=1/4$ cm.
- $r+r+r+r+r+r=1$ which gives $6r=1$ and thus $r=1/6$ cm

Next, we determine the areas of the circles in both scenarios:

- Case M:
 - Area of the four circles $=4\times\pi R^2$
- Case N:

- Area of the nine circles $= 9 \times \pi R^2$
- Area of the square in both cases $= 1 \text{ cm}^2$.

We can now calculate the areas of the unshaded regions:

- Case M:
 - Area of unshaded regions $= 1 - 4 \times \pi (1/4)^2 = 1 - \pi/4$
- Case N:
 - Area of unshaded regions $= 1 - 9 \times \pi (1/6)^2 = 1 - \pi/4$

Therefore, the ratio of the unshaded areas in Case M to that in Case N is 1:1.

34.) Solution: A

The hands of a clock align 11 times within a span of 12 hours (Considering that they only coincide once between 11 and 1, specifically at 12 o'clock). They overlap approximately every 65 minutes, rather than precisely every 60 minutes.

Visualizing it this way:

Starting from 3 PM: 3 PM $\rightarrow$ 4:05 PM $\rightarrow$ 5:10 PM $\rightarrow$ 6:15 PM $\rightarrow$ 7:20 PM

$\rightarrow$ 8:25 PM $\rightarrow$ 9:30 PM $\rightarrow$ 10:35 PM $\rightarrow$ 11:40 PM

$\rightarrow$ 12:45 AM $\rightarrow$ 1:50 AM $\rightarrow$ 2:55 AM

35.) SOLUTION: D

"Heal" pertains to improving in health, recuperating from an ailment, or recovering from an injury. On the other hand, the "heel" is the rear portion of one's foot and is also the section of socks and stockings that cover this part.

The primary distinction between "heal" and "heel," and a helpful way to differentiate between them, is that "heal" functions as a verb, whereas "heel" is a noun.

As one ages, a damage to the "heel" might require more time to "heal".

36.) SOLUTION: A

Given that,

- **P is rung every 10 minutes**
- **Q is rung every 20 minutes**
- **R is rung every 30 minutes**

Now, $\text{LCM}(10, 20, 30) = 60 \text{ minutes} = 1 \text{ hour}$.

If all the three bells are rung at 12:00PM, when will after that all the three bells ring together again the next time $= 12:00\text{PM} + 1 \text{ hours} = 1:00 \text{ PM}$.

37.) SOLUTION: A

- **Conclusion I: Some bottles are knives –True.**
- **Conclusion II: Some knives are cups –True.**
- **Conclusion III: All cups are bottles –False.**
- **Conclusion IV: All knives are cups –False.**

38.) SOLUTION B

	S	M
Cost price:	90x	100x
Delivery charge:	₹150	₹ 0

A person bought the items from the store S and saved ₹ 100.

$$\text{So, } 90x + 150 = 100x - 100$$

$$\Rightarrow -10x = -250$$

$$\Rightarrow x = 25$$

∴ The price of the item at the online store S, if there are no other charges includes $= 90x = 90 \times 25 = ₹ 2250$.

39.) SOLUTION: D

The passage states that controlling mosquitoes using chemicals may have undesired consequences.

However, it does not provide any information about whether using genetically modified mosquitoes has any negative consequences or not as made clear by this sentence: "It remains to be seen if this novel approach has unforeseen consequences"

Therefore, we cannot make a definitive judgment about the safety or potential negative consequences of genetically modified mosquitoes based on the information provided in the passage.

40.) SOLUTION C

Given that, the plan of the gallery where the boundary walls are opaque (not able to be seen through; not transparent).

If the security guard will be posted on 'P' then he/she can't watch the visibility side of 'R' and vice versa. But if the security guard will be posted at 'Q' and 'S' then the security guard can watch over the entire inner space of the gallery.

41.) SOLUTION B

Calculate manually

42.) SOLUTION B

Let '3w' and 'w' units be the work done by A and B respectively in a day and x be the total work in units. Then,

$$x/w - x/3w = 40$$

$$\Rightarrow x = 60w$$

$$\text{Hence required time} = 60w/(w+3w) = 15$$

43.) SOLUTION C

Let the number of fishes of type C be '8x' Number of fishes of type A = 10x

Number of fishes of type B = 9x According to the question,

$$9x = 144 \Rightarrow x = 16.$$

The total number of fishes of types A, B and C put together = 27x.

It's also known that the number of fishes of Type D constitutes 20% of the

total number of fishes.

$$\text{Hence, the total number of fishes in the pond} = (27x/4) * 5 = (27*16)/4 * 5 = 540$$

44.) Solution D

The most appropriate word which fits here is 'tally'. It gives the correct meaning of the sentence.

45.) Solution: (C)

Vexation meaning the state of being annoyed, frustrated, or worried

Splendid meaning magnificent; very impressive.

Joyful meaning feeling, expressing, or causing great pleasure and happiness.

Annoying means the same as Vexation.

Proud meaning feeling deep pleasure or satisfaction as a result of one's own achievements, qualities, or possessions or those of someone with whom one is closely associated.

So annoying the synonym for Vexation.

46.) SOLUTION B

$$\log (x - 13) + \log 2^3 = \log (3x + 1)$$

$$\Rightarrow \log (x - 13) + \log 8 = \log (3x + 1)$$

$$\Rightarrow \log (8x - 104) = \log (3x + 1)$$

$$\Rightarrow 8x - 104 = 3x + 1$$

$$\Rightarrow 5x = 105$$

$$\Rightarrow x = 21$$

47.) SOLUTION A

We can take the percentage here by using the degree directly.

$$\text{Percentage of students who passed in English} = (54/360) * 100 = 15\%$$

Please note that the total is 360degree in a pie chart.

48.) SOLUTION C

1, 5, 7 have two similar elements, one inside the other.

2, 4, 8 have one element placed inside a different element.

3, 6, 9 have two similar elements, one inside the other and the area between the two elements is shaded.

49.) SOLUTION D

The meaning of the word “hospitality” is the quality or disposition of receiving and treating guests and strangers in a warm, friendly, generous way which justifies that option (d) is the correct answer

50.) Solution B

We already know that the sum of n natural numbers is $n(n+1)/2$. So the sum of $2n$ natural numbers will be $(2n*(2n+1))/2=n(2n+1)=2n^2+n$. And the sum of the first n odd natural numbers is n^2 .

So difference $=2n^2+n-n^2=n^2+n$.

51.Solution B

Cost of multiplying $AB=4*2*2=16$

Cost of multiplying $CD=2*4*3=24$

Cost of multiplying $((AB)(CD))=4*2*3 + 12 + 24=16+24+24=64$

52.SOLUTION D

Initially $i=0, j=1, a=10$

Condition $a\%j == 0$ true, then it will call $\text{foo}(10+1)$, it returns 14.

$i = 14, j=2, a=10$

Condition $a\%j ==0$ true then it will call $\text{foo}(10+2)$, it returns 27

i= 41

Condition a % j == 0 false, final value of i = 41

53. SOLUTION C

In a bipartite graph, every cycle has even length, so in particular, every face has an even number of edges. A face cannot have only 2 edges, because there would be 11 double edges. A face with 6 edges can always be subdivided into two faces with 4 edges each {and the same is true for faces with even more edges}, so in any planar bipartite graph with a maximum number of edges, every face has length 4. Since every edge is used in two faces, we

have $4F = 2E$. Plugging this into Euler's formula, we find that $V + 1/2E = E + 2$, which we

rewrite as $E = 2V - 4$. Since $n = V$, the answer is $2n - 4$ edges

2nd method

$$\mathbf{V-E+R=2}$$

Sum of degrees of all regions=2E

So $4R=2E$ (EVERY REGION HAS 4 degree 4 in case of bipartite)

$$\mathbf{R=2E/4}$$

$$\mathbf{V-E+2(E/4) =2}$$

$$\mathbf{V-(E/2) =2}$$

$$\mathbf{2V-E=4}$$

$$\mathbf{2V-4=E}$$

54 SOLUTION C

Given that

Option A:- $f(n)=O(i(n))$ $f(n)\leq ci(n)$

Nothing can be said about the relation in between $i(n)$ and $f(n)$

Option B:- $i(n)=O(f(n))$, $i(n)\leq cf(n)$

Nothing can be said about the relation in between $i(n)$ and $f(n)$

Option C:- $i(n)=O(h(n))$, $i(n)\leq ch(n)$

So option C is correct

In option D it is given that $g(n)=O(i(n))$ $g(n)\leq c(i(n))$

But there is no relation between $g(n)$ and $i(n)$

55.) Solution D

The range of host in this subnet is 192.168.4.00 000001 to
192.168.4.00 111110

Which is 192.168.4.1 to 192.168.4.62

The option d is correct answer

56.) Solution C

Here both the statements are true because weak entities are dependent on the strong entities, thus, weak entities are deleted automatically on the deletion of its strong entity. Also, they are physically stored together.

57.) Solution: D

When using the delete cascade condition, upon deleting the tuple (1, 6), all tuples associated with value A = 1 are removed. This includes tuples (5, 1) and (2, 1). Consequently, 2 tuples are deleted in this step.

Subsequently, deleting (5, 1) and (2, 1) leads to the removal of

tuples referencing A values of 2 and 5. This results in the deletion of tuples (19, 2) and (10, 2). Therefore, a total of 4 tuples are deleted at this stage.

Given that no other tuple is linked to 19 and 10, no further deletions occur. Thus, the correct count of deleted tuples is 4.

58.) Solution D

The UDP header has four parts, each of two bytes. That means we get the following interpretation of the header.

Source port number = 0632

Destination port number = 000D

Total length = 001D

Converting 001D hexadecimal into decimal.

0000 0000 0001 1101 = 29 bytes

59.) Solution D

1st fragment: 4448 +20 bytes

2nd fragment: 4448 +20 bytes

3rd fragment: 1104 +20 bytes

MF= 1 because more fragments are there after this.

60.) Solution C

Initially, the system could handle a maximum of X client requests. Subsequently, an additional Y client requests were permitted, making the total allowable requests X + Y. By setting the initial value of the counting semaphore to X + Y, the server process can accommodate this total number of client requests.

61.) SOLUTION A

Ans ABAABA

62.) Solution B

1. Option A is true
2. Option B is false Port number of POP3 is 110.
3. Option C is true Yes; the client first sends the query to the DNS resolver which helps to get the IP by contacting the further top hierarchies.
4. Option D is true; IMAP is more powerful and contains more features than POP3.

63.) SOLUTION A

Let A_1 be the event of selectors, a win having heads both sides and A_2 be its complement.

Let E be the event of getting head when the selected coin is tossed.

$$P(A_1) = 2/16, P(A_2) = 14/16, P(E|A_1) = 1, P(E|A_2) = 1/2$$

$$P(E) = P(A_1) P(E|A_1) + P(A_2) P(E|A_2) = (2/16 * 1) + (14/16 * (1/2)) = 9/16.$$

64.) SOLUTION C

1, 0, 3, 5, 4, 2, 6 is an invalid BFS sequence because 5 is visited before vertices 4 and 2 as vertex 5 is at distance 2 and 4 and 2 are at distance 1 from the source (i.e. vertex 0).

65.) SOLUTION B

$|x-y|>5$ which means $x-y<-5$ or $x-y>5$

$x-y<-5$ $(x,y) \in \{(0,6), (0,7), (0,8), (0,9), (0,10), (1,7), (1,8), (1,9), (1,10), (2,8), (2,9), (2,10), (3,9), (3,10), (4,10)\}$

Similarly for $x-y>5$ we have 15 more examples

$$n(A) = 15 + 15 = 30, n(S) = 11 \times 11$$

$$P(A) = 30 / 121.$$

66.) SOLUTION A

For the given expression minimized expression of complement of $(BC' + A'D)(AB' + CD')$ will be:

$$\Rightarrow (B' + C)(A + D') + (A' + B)(C' + D)$$

$$\Rightarrow (B' + C + A' + B)(B' + C + C' + D)(A + D' + A' + B)(A + D' + C' + D)$$

$$\Rightarrow 1 \cdot 1 \cdot 1 \cdot 1 // B + B' = 1, C + C' = 1, A + A' = 1, D + D' = 1$$

Hence, it will have no literal.

67.) Solution B

In the function m,

$x=8$ will change the global variable.

void n will run and print 8

Main printf will print its local x which is 3

68.) Solution B

First transmission: 2 KB

Second transmission: 4 KB

Third transmission: 8 KB

Forth transmission: 16 KB

Fifth transmission: 20 KB (Threshold reached)

Sixth transmission: 22 KB

Seventh transmission: 24 KB

Eighth transmission: 26 KB

Ninth transmission: 28 KB (Time out occurs, New threshold = $28/2 = 14$ KB)

Tenth transmission: 2 KB

Eleventh transmission: 4 KB

Twelfth transmission: 8 KB

Thirteenth transmission: 14 KB (Th reached)

Fourteenth transmission: 16 KB

It takes 4 round trip time after the timeout to reach the 16 KB = 4×10 msec = 40 msec

69.) SOLUTION D

The number given is (12121) and is in base 3.

2's complement means that we need to find the (r-1)'s complement of (12121).

=> $(3-1) = 2$'s complement we will subtract 12121 from 22222 = $22222 - 12121 = (10101)_3$

=> (10101) in base 2 will be = 01011011_2 .

70.) SOLUTION B

For a tautology $P \square Q$ i.e $P' + Q = 1$ ALWAYS

I. Not a tautology

II. Is a tautology.

III. Is not a tautology as $P \square Q$ is not 1 always

71.) SOLUTION D

Explanation:

The MD5 message-digest hashing algorithm processes data in 512-bit blocks.

72.) SOLUTION A

The head moves in one direction, either to the center or to the outermost track, servicing requests as it passes the corresponding tracks. It then turns around and repeats this in the other direction.

72,56,50,32

73.) Solution C

Output = $C'D + CD'$

So, 0110 should be on the input lines

74.) SOLUTION A

PUSH R: Store or push the content of the register R into stack memory. Content is always pushed on top of the stack.

POP R: Read/Pop the data from the top of the stack memory into register R.

Content is always popped from the top of the stack.

75.) SOLUTION A

This is a prefix code as no code word is the prefix of any other. No, this would never be generated by Huffman's algorithm. There would be no reason for generating the codeword 101, since the shorter code 10 would suffice.

76.) SOLUTION D

S1 has unique solution

S2 has no solution

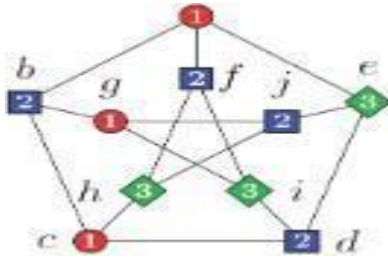
77.) SOLUTION B

Solution

G has an odd cycle, so cannot be bipartite, i.e. $\chi(G) > 2$

By putting 3 colors, we get the solution:

G



Chromatic no = 3

78 SOLUTION A

Take $t = (x^2 + 2x + 3)^{1/2}$

$$dt = x + 1 / ((x^2 + 2x + 3)^{1/2})$$

$$\sqrt{x^2 + 2x + 3} + c$$

79.) Solution C

Here the first element is pivot. If we run the partition algorithm one

time, we will get the following result 6 9 7 5 11 15 13 12 17 18

80.) SOLUTION A

Solution: 112 keys

Since there are 3 candidate keys, therefore, the number of super keys will

$$\text{be: } 2^{n-1} + 2^{n-1} + 2^{n-1} - 2^{n-2} - 2^{n-2} - 2^{n-2} + 2^{n-3}$$

Since $n = 7$, therefore,

$$\Rightarrow 2^6 + 2^6 + 2^6 - 2^5 - 2^5 - 2^5 + 2^4$$

$$\Rightarrow 64 * 3 - 96 + 16$$

$$\Rightarrow 192 - 80$$

$$\Rightarrow 112 \text{ super keys.}$$

81.) SOLUTION c

I. False – with RR, identical jobs will all finish at nearly the same time (at the very end of the workload time), which has a very poor average turnaround time.

II. False - To implement it we need to know runtime of all jobs in advance which is not possible in practice

III. True - Since RR scheduling gives slots for every job for fixed time quantum hence starvation is not possible.

82.) SOLUTION B

The binary representation for the given 32 bit hexadecimal value is 11110001111000001111101001100110

Places where value is 1=18

So places where the track is empty = $32-18=14$

Percentage of empty track = $(14/32) \times 100 = 43.75$.

83.) Solution B

points to 3.

$p += 2$; will make p point to 5

$*p += 2$; will make $a[5]$ as 7.

$*p++$; here postfix $++$ has higher precedence so we will do $p++$ first.

But since it is postfix hence result will not be reflected in same line.

Now we do $*p$ which is same as 7.

In the next line, p points to $a[6]$

84.) Solution: D

Explanation:

Checksum and TTL is updated by the routers while forwarding the packet.

85.) Solution B

The comma operator with bracket moves from left to right and return the rightmost value

If condition is true

$i = (7, 4, 2) // 2$ will return

The value of $i=2$

86.)Solution B

$$1-(0.6)^3=0.784.$$

87.) Solution C

The MST of the above graph can be determined by Prim's or Kruskal's algorithm; the MST consists of the edges AB, BC, CD, DE, DF, the cost is given by $2+2+2+3+3=12$.

88.) Solution D

Everything is possible in the order of constant time, in the circular doubly linked list we have the flexibility to move the head pointer to the last of the linked list in the $O(1)$.

89.) Solution D

$$A'BC + AB'C' + A'B'C' + AB'C + ABC$$

$$=A'BC + AB'C' + AB'C' + A'B'C' + AB'C + ABC \quad (AB'C' = AB'C' + AB'C')$$

$$=A'BC + ABC + A'B'C' + AB'C' + AB'C' + AB'C$$

$$=BC(A' + A) + B'C'(A' + A) + AB'(C' + C)$$

$$=BC + B'C' + AB'$$

Hence, Option(D) $AB' + BC + B'C'$ is the correct choice.

90.) Solution D

$$(292)_{10} = (1204)_x$$

$$2 \cdot 10^2 + 9 \cdot 10 + 2 = x^3 + 2 \cdot x^2 + 0 + 4$$

$$200 + 90 + 2 = x^3 + 2x^2 + 4$$

$$x^3 + 2x^2 - 288 = 0$$

solution for $x = 6$

None of the options satisfy the equation.

91. Solution A

There are two Candidate keys ABC and ABD. A proper subset of some candidate key can determine a non-prime attribute, for example, BC which is the proper subset of ABC, and taking the closure of BC will give you BCDE, and as E is part of the closure of BC it means the non-prime attribute (E) can be determined by a proper subset of some Candidate key so it will be the violation of 2NF. if not in 2NF then higher NF also not possible.

92.) SOLUTION B

Option B: $YZ \rightarrow X$

for every value of YZ we can get value of X so its valid

93.) SOLUTION B

TRUE, Primary constraints are Not NULL and it should be unique key

FALSE, SQL has more power than Relational Algebra because basic relational algebra query cannot perform mathematical aggregate functions

94.) Solution D

The queries Q1 and Q2 will produce different results. The number of attributes in the resultant set of Q1 will be 3 which are (a, b, c) while the number of attributes in the resultant set of Q2 will be 4 (a, R.b, S.b, c).

Thus the result of Q1 and Q2 will not be the same. Hence, the correct option is (iv).

95.) SOLUTION C

Solution: 4

For entity E1, E2 we need only 1 relation (due to one-to-one relationship in between them) While a separate relation is required for entity E3. E3 could be merged with E1 but cannot be merged with E2 due to a one-to-many relationship with total participation on many sides. Thus, 2 relations are required for E1E2 and for E3 respectively. Again, 2 separate relations are required for attributes A and B. Therefore, the total number of relations satisfying 1NF is 4.

96.) Solution A

Applying column transformations

$$C2 \rightarrow C2 - C1$$

$$C3 \rightarrow C3 - C1$$

$$C3 \rightarrow C3 - C2$$

We get a matrix having C2 and C3 as identical so $|A|=0$

97.) SOLUTION B

Explanation:

If x is 0, we have a deadlock immediately.

If x is 1, process D can run to completion.

When it is finished, the available vector is 1 1 2 2 1.

Now A can run to complete, the available vector then becomes 2 1 4 3 2.

Then C can run and finish, return the available vector as 3 2 4 4 2.

Then B can run to complete. Safe sequence D A C B.

98.) SOLUTION D

In one minute, CPU is busy = $10 \times 5 = 50$ sec out of 60 sec. So, fraction of the time CPU is ideal is $60 - 50 = 10$ sec out of 60 sec, then in percent $10/60 = 16.67\%$

99.) Solution D

Solution:

Push A = s: A(top)

Enqueue B = q: B(head)

Push C = s: A C(top)

Pop = top element C is deleted from stack then s:A(top)

Pop = A is deleted from stack then stack will become empty

Enqueue D = q: B(head), D

Push E = s: E(top) → 1

Dequeue = B is deleted from queue then q: D(head)

Enqueue F = q: D(head) F → 2

At last, there will be 3 elements left. So, answer is 3.

So, the correct answer is 3.

100.)Solution D

Explanation:

MF= 1 means it is an intermediate fragment. If a fragment is intermediate, it means that the fragment size is similar to the previous fragments.

MTU is 500 bytes The maximum allowed size of the packet is **480 + 20 bytes header**

First fragment offset number will be = $480/8 = 0 - 59$

Second fragment offset number will be= $480/8 = 60 - 119$

Third Fragment offset number will be = $480/8 = 120 - 179$

Fourth fragment offset number will be = $480/8 = 180 - 239$